

AI-Based Pricing Model for Used Cars in India: A Data-Driven Analysis of Feature Importance, Model Performance, and Business Impact

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March 2026

Abstract

This paper presents a machine learning-based framework for predicting used car prices in the Indian market. The pricing of used vehicles is a structurally complex regression problem shaped by both objective technical characteristics and market-driven effects such as brand perception, model positioning, and location. To address this problem, a structured preprocessing and model benchmarking workflow was developed using a dataset of 7,253 used car listings containing numerical and categorical vehicle attributes. The target variable was modeled on a logarithmic scale to stabilize variance and improve generalization.

Multiple regression models were trained and compared, including Linear Regression, Decision Tree, Tuned Decision Tree, Lasso Regression, and Random Forest. The results show that Random Forest provides the best overall predictive performance, achieving approximately $R^2 \approx 0.91$ on the test data with the lowest root mean squared error. The analysis further shows that price formation follows a dual structure: technical variables such as power and vehicle age define the base economic value of a vehicle, while market-related variables act as secondary adjustments. Feature importance analysis confirms that a small set of dominant predictors explains a large share of observed price variation.

Beyond model accuracy, the paper provides a business-oriented interpretation of the findings and quantifies the economic value of improved pricing quality. The proposed approach constitutes a scalable and practically relevant foundation for AI-driven pricing systems in fragmented automotive markets.

1 Introduction

The used car market is one of the most dynamic segments of the automotive industry. In contrast to new vehicle pricing, which is usually anchored by manufacturer list prices and structured discount systems, the valuation of used vehicles is characterized by heterogeneity, uncertainty, and substantial information asymmetry. Each used car differs with respect to age, mileage, condition, powertrain, ownership history, and perceived desirability. Consequently, pricing decisions are often inconsistent across sellers, regions, and market channels.

This challenge is particularly relevant in the Indian used car market, where digital platforms, fragmented inventories, and wide price dispersion create a strong need for scalable pricing mechanisms. In such an environment, inaccurate pricing can generate significant economic losses. Underpricing directly reduces margin, while overpricing increases holding costs and slows inventory turnover. A data-driven pricing model can improve both consistency and accuracy while providing decision support for dealerships, platforms, and financial institutions.

This paper investigates the extent to which used car prices can be modeled through structured machine learning methods. The study compares linear and nonlinear regression models and examines the relative role of technical and market-oriented drivers. In contrast to purely descriptive benchmarking, the objective here is to derive a predictive pricing function and interpret its structure from both a data science and business perspective.

The main contributions of this paper are as follows:

1. Development of a structured machine learning workflow for used car price prediction in the Indian market.
2. Comparative evaluation of multiple regression models with respect to predictive performance and generalization.
3. Identification of dominant price drivers using feature importance analysis.
4. Derivation of a dual-structure interpretation of price formation combining technical base value and market adjustment effects.
5. Quantification of the practical business impact of improved pricing accuracy.

2 Related Work

Used car price prediction has been studied using a broad spectrum of statistical and machine learning methods. Existing work typically compares linear models, tree-based methods, and, in some cases, neural-network-based approaches. Prior studies show that Random Forest and other nonlinear methods often outperform classical linear models in heterogeneous automotive datasets,

especially when price formation depends on feature interactions and nonlinear depreciation effects.

Recent studies on the Indian used car market also emphasize the value of preprocessing, outlier treatment, and categorical encoding. Ensemble methods have been shown to provide strong robustness in the presence of noisy real-world data. However, many published studies remain primarily application-oriented and stop at reporting model metrics. They provide limited interpretation of the economic structure of price formation and often do not explicitly separate technical and market-related drivers.

The present study extends this line of work by combining predictive modeling with structural interpretation. Instead of asking only which model performs best, this paper also asks why certain features dominate, how price drivers interact, and what the findings imply for real-world pricing systems.

3 Problem Formulation

From a machine learning perspective, the task is formulated as a supervised regression problem. Let

$$X \in \mathbb{R}^{n \times p} \quad (1)$$

be the feature matrix with n observations and p explanatory variables, and let

$$y \in \mathbb{R}^n \quad (2)$$

be the target vector containing used car prices. The objective is to learn a function

$$\hat{y} = f(X) \quad (3)$$

that minimizes prediction error on unseen data.

Because raw vehicle prices exhibit right-skewness and heteroscedasticity, the target was transformed using a logarithmic function. The effective regression target is therefore

$$y^* = \log(\text{Price}). \quad (4)$$

This transformation improves numerical stability and reduces the disproportionate influence of high-price outliers.

From a business perspective, the pricing mechanism can be interpreted as the superposition of two partially distinct components:

$$\text{Price} = f_{\text{tech}}(X_{\text{tech}}) + f_{\text{market}}(X_{\text{market}}) + \varepsilon, \quad (5)$$

where $f_{\text{tech}}(\cdot)$ captures the technical or objective base value of the vehicle, while $f_{\text{market}}(\cdot)$ captures adjustment effects related to market perception. This dual structure serves as a central interpretive lens throughout the paper.

4 Dataset and Preprocessing

4.1 Dataset Description

The dataset used in this study contains 7,253 used car listings and 14 variables describing both technical and market-related vehicle characteristics. The features include numerical variables such as power, year, mileage, engine capacity, kilometers driven, seats, and price, as well as categorical variables such as brand, model, transmission, location, fuel type, and owner type. The target variable is the used car selling price measured in lakh INR.

The dataset spans a broad range of market segments, from low-budget vehicles to premium cars. This heterogeneity makes the prediction task economically relevant but statistically challenging, because the same feature can have different marginal effects across segments.

4.2 Preprocessing Strategy

Several preprocessing steps were carried out before model training:

1. removal of irrelevant identifiers,
2. cleaning and consistency checks for numerical variables,
3. selective missing value imputation,
4. encoding of categorical variables,
5. log-transformation of the target variable.

A particularly important preprocessing decision concerned the feature *New Price*. Although conceptually attractive as a proxy for original vehicle value, this variable contained more than 6,000 missing observations and was therefore removed. This decision reflects a general modeling principle: theoretical relevance does not justify inclusion when data quality is insufficient for reliable inference.

4.3 Missing Value Treatment

Numerical variables such as *Engine*, *Power*, and *Mileage* contained missing values and were imputed using grouped median strategies. For *Engine* and *Power*, a hierarchical imputation logic was applied using *Brand + Model*, then *Brand*, and finally the global median as fallback. For *Mileage*, grouped imputation by fuel type was applied. Median-based imputation was selected in order to reduce sensitivity to outliers.

Categorical variables were complete and required no imputation. The overall preprocessing design deliberately prioritized robustness and interpretability over aggressive feature reconstruction.

5 Feature Engineering and Feature Space

The feature engineering process was designed to represent the economic state of each vehicle as faithfully as possible while preserving model tractability. A central derived variable is vehicle age:

$$Age = CurrentYear - Year. \quad (6)$$

This variable provides a more interpretable representation of depreciation than the raw production year.

Categorical variables were transformed into a numerical feature space using one-hot encoding. If c_j denotes a categorical level, then the encoded representation becomes a binary indicator variable

$$z_j = \begin{cases} 1, & \text{if category is present,} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

This expands the feature space while preserving categorical identity. At the same time, it distributes the influence of a market factor such as brand across multiple binary dimensions rather than concentrating it in a single variable.

The resulting feature vector for a vehicle can therefore be expressed as

$$x_i = [x_i^{tech}, x_i^{market}], \quad (8)$$

where x_i^{tech} contains technical variables and x_i^{market} contains encoded market variables. This decomposition aligns naturally with the dual price-formation structure introduced above.

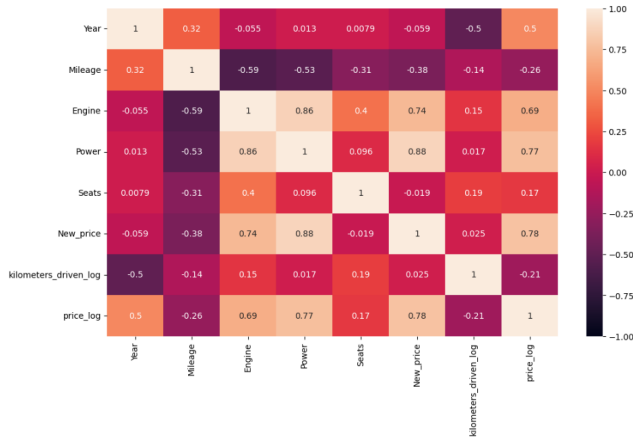


Figure 1: Correlation heatmap for the main numerical variables.

6 Methodology

6.1 Model Benchmarking Framework

To compare predictive performance across levels of model complexity, five regression models were trained and evaluated:

1. Linear Regression,
2. Decision Tree,
3. Tuned Decision Tree,
4. Lasso Regression,
5. Random Forest.

These models span a spectrum from highly interpretable linear structure to nonlinear ensemble learning. The purpose of the comparison is not merely to identify the numerically best model, but also to understand which structural assumptions are most appropriate for the pricing problem.

6.2 Linear Regression

Linear Regression assumes a linear relationship between the feature matrix and the target:

$$y = X\beta + \varepsilon, \quad (9)$$

where β is the coefficient vector and ε is the residual error term. The coefficient estimates are obtained by minimizing the residual sum of squares:

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (10)$$

This model provides a useful baseline and is particularly valuable for interpretation. If a linear model performs reasonably well, then a substantial part of the market can be explained through approximately additive effects.

6.3 Lasso Regression

Lasso Regression augments the linear objective with an L_1 penalty:

$$\hat{\beta} = \arg \min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]. \quad (11)$$

This regularization encourages sparsity and can reduce model complexity, but it may also suppress useful variables when correlated predictors are present.

6.4 Decision Tree

Decision Trees recursively partition the feature space into subregions and assign constant predictions within each region. They can naturally model nonlinearities and interactions, but unconstrained trees are prone to overfitting.

6.5 Random Forest

Random Forest addresses overfitting by aggregating many decorrelated decision trees. The ensemble prediction is defined as

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x), \quad (12)$$

where T is the number of trees and $f_t(x)$ is the prediction of tree t . Because each tree is trained on a bootstrap sample and each split considers only a subset of features, Random Forest can capture nonlinear interactions while maintaining better generalization than a single tree.

7 Experimental Design

The dataset was split into training and test subsets. The training data were used for fitting the candidate models, while the test data were reserved for out-of-sample evaluation. Hyperparameter tuning was applied for the tuned Decision Tree and Random Forest using cross-validation.

Model performance was evaluated using the coefficient of determination R^2 and the root mean squared error (RMSE):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (13)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (14)$$

The evaluation logic follows two principles. First, the best model must minimize out-of-sample error rather than merely fit the training data. Second, performance must be interpreted jointly with structural behavior such as overfitting, stability, and interpretability.

8 Results

8.1 Model Comparison

Table 1 summarizes the central model results.

The results clearly indicate that Random Forest provides the best overall predictive performance. It achieves the highest test R^2 and the lowest test RMSE while maintaining a moderate gap between training and test performance. The standard Decision Tree, by contrast, achieves perfect training fit but materially worse test behavior, which is a textbook overfitting pattern.

8.2 Interpretation of Model Behavior

Three conclusions can be drawn from the model comparison. First, a large share of price variation can already be explained through relatively simple linear relations, as shown by the strong baseline performance of Linear Regression. Second, the superiority of Random Forest confirms the presence of nonlinear interactions and threshold effects in the pricing mechanism. Third, model tuning materially improves tree-based generalization, showing that the bias-variance tradeoff is central to this application.

8.3 Segment-Level Error Behavior

The segment analysis showed strong performance for low-value and typical cars, while materially larger deviations were observed in the premium segment. This pattern suggests that premium prices are influenced by additional latent factors not fully captured in the current feature space, such as trim level, condition, or localized demand shocks.

9 Feature Importance Analysis

A key result of the project is the feature importance structure derived from the Random Forest model. The feature importance distribution reveals a highly concentrated structure, where a small subset of variables explains a large share of the total variance in used car prices. In particular, *Power* emerges as the most influential predictor, followed by *Year* or *Age*, depending on the exact feature representation. This indicates that vehicle performance class and depreciation dynamics form the economic backbone of price formation.

Interestingly, *Kilometers Driven* plays a less dominant role than might be expected. This suggests that its informational content is partially absorbed by correlated features such as age. From a modeling perspective, this confirms that the Random Forest prioritizes the most informative representation of vehicle usage rather than assigning redundant importance across correlated variables.

Market-related features such as brand and location appear with lower individual importance scores. However, due to one-hot encoding, their influence is distributed across multiple binary variables, which reduces their apparent individual impact while preserving their aggregate effect.

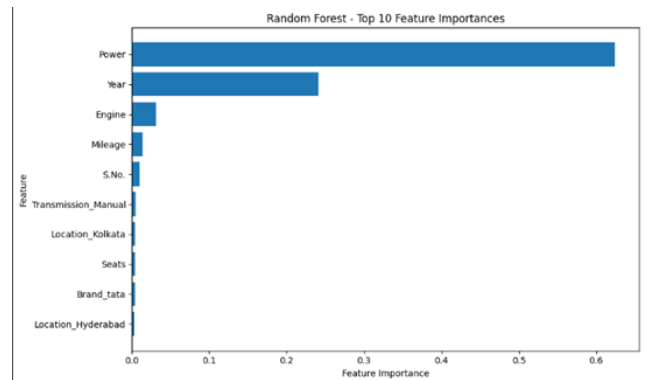


Figure 2: Feature importance derived from the Random Forest model. Power and age emerge as the dominant predictors of used car prices, while mileage-related variables show reduced importance due to correlation effects.

Table 1: Comparison of model performance

Model	Train R^2	Test R^2	Train RMSE	Test RMSE
Linear Regression	0.981	0.866	1.574	3.994
Decision Tree	1.000	0.875	0.000	3.858
Tuned Decision Tree	0.948	0.894	2.573	3.550
Lasso Regression	0.820	0.884	4.797	3.710
Random Forest	0.976	0.913	1.743	3.205

10 Error Analysis and Diagnostics

A strong paper should not stop at average performance metrics. It should also examine where the model fails. The segment analysis indicates that the model performs well for low and typical market segments, while deviations increase for premium vehicles. This implies that residual variance is not constant across the market distribution.

A plausible explanation is that premium prices are influenced by omitted variables such as brand prestige, special editions, condition quality, or local scarcity effects. These factors are only partially captured in the current dataset. This observation is consistent with the broader principle that model error tends to increase when unobserved heterogeneity increases.

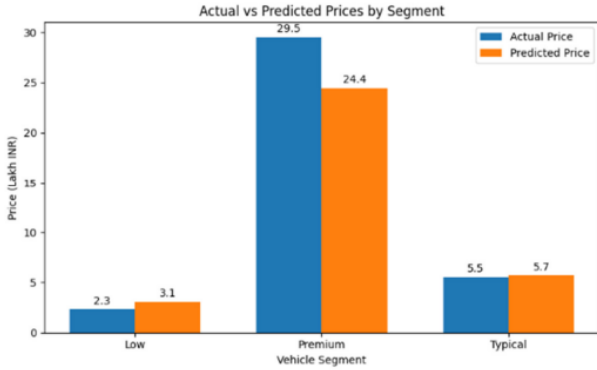


Figure 3: Predicted vs. actual values for the Random Forest model. Most observations are closely aligned with the diagonal, indicating strong predictive accuracy. Deviations are more pronounced in the premium segment.

The predicted versus actual plot confirms the strong overall performance of the Random Forest model. Most observations are closely aligned with the diagonal line, indicating that predicted values closely match actual market prices. However, deviations become more pronounced in the upper price range. This suggests that premium vehicles are influenced by additional latent variables not fully captured in the dataset, such as condition, trim level, or localized demand effects.

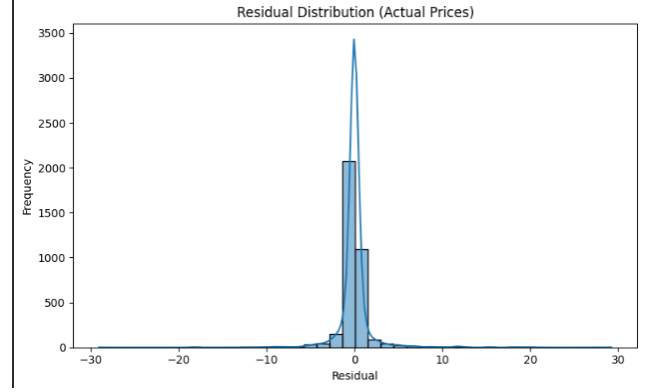


Figure 4: Residual distribution of the Random Forest model. Most residuals are concentrated around zero, indicating good overall fit. Slight asymmetry suggests mild underprediction bias.

The residual distribution is centered around zero, which indicates that the model does not exhibit strong systematic bias. However, a slight asymmetry towards negative residuals suggests a tendency to underpredict higher-priced vehicles. This observation aligns with the deviations seen in the premium segment and reinforces the need for additional features or segmentation strategies.

11 Discussion

The results indicate that used car price formation in the Indian market follows a dual structure. Technical attributes define the underlying economic value of the vehicle, whereas market-related factors act as secondary adjustments. This finding is important because it connects predictive modeling with an interpretable business logic.

In formal terms, the results support the decomposition

$$Price \approx f_{tech}(Power, Age, Engine, Mileage) + f_{market}(Brand, Model, Location) \quad (15)$$

The Random Forest model is particularly effective because it can represent both dominant drivers and interaction effects without requiring the analyst to specify them manually in advance.

The results suggest that used car pricing is not purely a statistical problem, but rather an economic valuation problem with a measurable structure. Machine learning models such as Random Forest do not only approximate

price, but implicitly learn the underlying pricing logic of the market.

The model comparison also reveals an important methodological lesson. Simpler models such as Linear Regression remain valuable because they provide interpretability and a robust baseline. However, a heterogeneous market with nonlinear depreciation and interaction effects requires a more flexible model for operational deployment. Random Forest offers the best balance between predictive power, robustness, and acceptable interpretability.

This finding is particularly relevant for practical applications in cost engineering and pricing strategy, where understanding the structure of price drivers is as important as predictive accuracy itself.

12 Business Impact and Implementation Relevance

The practical relevance of improved pricing accuracy is substantial. Based on the project analysis, the average vehicle price in the dataset is approximately 9.48 lakh INR. A pricing improvement of just 1% therefore corresponds to approximately 9,480 INR per vehicle. For a mid-sized dealer handling 1,000 to 1,500 vehicles annually, this translates into an annual impact of roughly 9.5 to 14.2 million INR.

This result is significant because it shows that even modest metric improvements can create economically meaningful value at scale. The model can be integrated into dealer management systems or platform-based valuation tools to generate standardized price recommendations. Such deployment can reduce reliance on subjective judgment, improve pricing consistency, and support faster inventory decisions.

From an implementation perspective, the current model already constitutes a viable global pricing model. A natural next step is the introduction of segmentation logic, where vehicles are first grouped into more homogeneous clusters and only then modeled with local pricing functions. This would likely improve premium-segment performance further.

13 Limitations

Despite the strong results, the present study has several limitations. First, the model is constrained by data quality and feature completeness. Important variables such as vehicle condition, accident history, trim level, refurbishment status, and demand-side intensity are not explicitly observed. Second, the removal of *New Price* was methodologically justified, but it also reduced access to a potentially strong baseline valuation feature. Third, one-hot encoding distributes the influence of certain market factors across multiple variables, which reduces direct interpretability of their aggregate effect. In addition, the premium segment analysis suggests that

model uncertainty increases in higher-priced submarkets. This indicates that a single global model may not be fully sufficient across all market strata. Finally, the current study focuses primarily on supervised regression accuracy. A more advanced production system would also require continuous monitoring, drift detection, and retraining logic.

14 Future Work

Several natural extensions follow from this work:

1. develop segment-specific models for low, typical, and premium vehicles,
2. integrate additional features such as condition data or regional pricing signals,
3. add learning-curve experiments to quantify the marginal value of additional data,
4. compare Random Forest against gradient boosting methods such as XGBoost or LightGBM,
5. add explainability techniques such as SHAP for local and global interpretation,
6. deploy the pricing model in a lightweight application for real-time valuation.

15 Conclusion

This paper presented a machine learning-based approach for used car price prediction in the Indian market and evaluated multiple regression models with respect to performance, robustness, and interpretability. The Random Forest model emerged as the preferred solution, achieving the strongest out-of-sample performance and capturing the nonlinear structure of the pricing problem more effectively than linear baselines.

Beyond raw prediction metrics, the analysis showed that price formation follows a dual structure. Technical characteristics such as power and age define the base value of a vehicle, while market-related variables act as secondary adjustment layers. This interpretation connects the statistical model to a realistic economic logic and strengthens the case for implementation.

The study therefore contributes not only a predictive model, but also a structured way of thinking about automotive price formation in heterogeneous markets. In that sense, it represents a step toward broader AI-driven decision systems for pricing, risk assessment, and market optimization.

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